Sahaj Bhadja

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Summary

Full-stack Software Engineer pursuing an MSCS at USC with 2+ years building customer-facing web products, application-layer APIs, authentication, and dashboards in TypeScript, React, and Python, with hands-on experience translating AI and computer-vision outputs into actionable user experiences.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **full-stack features and backend services** on Azure SQL with row-level authorization and normalized data modeling for 500+ users, cutting dashboard latency from 1.2s to 350ms via query optimization.
- Owned **application-layer APIs** and async job orchestration, troubleshooting data patterns across system boundaries and shipping CI/CD-backed releases at 99%+ production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built a **React dashboard and Flask APIs** translating multimodal AI and vision-model outputs into clear, actionable user experiences for 150+ users handling 10K+ daily requests.
- Triaged issues spanning **application and model context**—mislabeled signals and missing detections—across switchable LLM and vision providers, validating customer-facing behavior under peak load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Collaborated with perception research to **interpret computer-vision model outputs**, building data pipelines over 100K+ trajectory logs to ensure the right signals powered downstream planning features.
- Integrated HD-map geometry and **AI outputs** via attention-based encoders, improving planning accuracy **18%** and surfacing edge cases and unexpected data patterns through closed-loop evaluation.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **full-stack MERN features and REST APIs** across eight microservices with JWT authentication and role-based access, syncing 2M+ records with real-time WebSocket dashboard updates.
- Automated Docker and Jenkins **CI/CD**, cutting deployment time **50%**, and used analytics-driven insights to reduce resource wastage **42%** with Sentry-based monitoring.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built **cross-platform React Native and React/TypeScript** clients with Flask APIs for 50K+ booking records, delivering sub-200ms responses and real-time Firebase synchronization.
- Integrated **secure payment and webhook workflows** with token-based auth, reducing manual handling **70%** and improving cross-platform compatibility **15%**.

Skills

Languages: TypeScript, JavaScript, Python, Java, C++, SQL
Frontend: React, React Native, Next.js, TailwindCSS, HTML/CSS, Responsive Dashboards
Backend & APIs: Node.js, Express, FastAPI, Flask, REST APIs, WebSockets, Microservices
Application Layer: Authentication, Authorization (RBAC/RLS), Data Modeling, JWT, Access Control
Data & AI: PostgreSQL, MongoDB, Azure SQL, PyTorch, Computer Vision, LLM/Multimodal Outputs, Time-Series Data
Practices & Tools: Git, CI/CD, GitHub Actions, Docker, Unit Testing, Debugging, Agile

Projects

PiXMod Live | *Next.js, CLIP, SAM, Stable Diffusion*

- Built an **AI computer-vision SaaS platform** using SAM and CLIP for image segmentation and semantic search across 100K+ embeddings, with a Next.js dashboard and Stripe subscription workflows.

FinNexus Live | *Next.js, TypeScript, Kubernetes, WebSockets*

- Built a **full-stack analytics platform** for 50K+ users and 1M+ transactions with real-time WebSocket updates, token-based access control, and Kubernetes-backed CI/CD.

HMIS Live | *MERN, TypeScript, Sentry*

- Built an **enterprise healthcare dashboard** (MERN, TypeScript) with consent-based access, appointment workflows, Twilio alerts, and Sentry monitoring, reducing administrative workload **30%**.